

returns—exactly what we see in data. In CAPM parlance, the voracious demand of leverage-constrained investors for high-beta stocks flattens the security market line (see chapter 6). The leverage constraint story, however, does not explain the underpricing of low-beta stocks relative to the market, only the overpricing of high-beta stocks. Thus, it cannot explain why low-beta or low-volatility assets have higher returns than the market portfolio, but it can explain why some low-beta assets have positive alphas. This story also predicts that leverage-constrained institutions should be attracted to high-risk stocks. In reality, though, institutional investors tend to underweight high-risk stocks; stocks with high idiosyncratic volatility are predominantly held and traded by retail investors.³¹

Agency Problems

Many institutional managers can't or won't play the risk anomaly. In particular, the use of market-weighted benchmarks itself may lead to the low volatility anomaly.³²

Figure 10.14 draws a theoretical relation between beta and expected returns in the diagonal solid line marked “CAPM” (the security market line). The data relation between returns and beta is the horizontal line marked “Data.” Consider stock A, which has positive alpha, and B, which has negative alpha. Unconstrained investors simply buy low and sell high. They buy A, which offers a high return relative to the CAPM, and they sell B, whose return is too low relative to the CAPM.

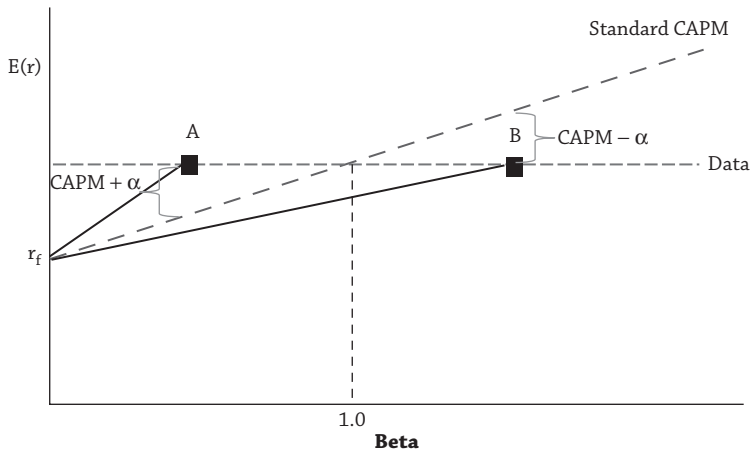


Figure 10.14

³¹ For an academic reference, see Han and Kumar (2011). Taking 13-F filings as of June 30, 2012 on Russell 1000 holdings, Martingale calculates that institutions hold 46.5% low-risk stocks and 53.5% high-risk stocks compared to a balanced 50%/50% split.

³² See Greenwood et al. (2010) and Baker, Bradley, and Wurgler (2011).